MEMORANDUM ESE5360
NANOFABRICATION AND NANOCHARACTERIZATION

Date: April 20, 2026
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Subject: Characterization of 2D Graphene Transistors as pH Sensors

Executive Summary

This report details the fabrication and characterization of two-dimensional (2D) graphene field-effect transistors (GFETs) utilized as microfluidic pH sensors. Device fabrication employed maskless photolithography utilizing a lift-off resist bilayer to define Cr/Au source and drain microelectrodes. Subsequently, a PMMA-assisted electrochemical delamination protocol was utilized to integrate chemical vapor deposition (CVD) grown monolayer graphene over the contacts. A polydimethylsiloxane (PDMS) microfluidic delivery system, fabricated via soft lithography, was bonded over the active channels to facilitate controlled analyte exposure.

Electrical transport measurements confirmed the structural integrity of the transferred graphene through linear resistance scaling. While static characterization revealed that baseline devices exhibited native p-doping due to environmental adsorbates and residual transfer polymers, process optimization yielded a champion device exhibiting a near-ideal Dirac point precisely at 0 V. Dynamic testing validated the fundamental pH sensing mechanism; devices exhibited distinct lateral shifts in their transfer characteristics when exposed to varying pH buffer solutions, confirming the electrostatic gating effect of the interfacial electrical double layer. Furthermore, power-law scaling analysis isolated the limiting effects of parasitic contact resistance on sensitivity. Overall, the GFETs successfully demonstrated highly sensitive, label-free biochemical detection, highlighting their viability for advanced microfluidic integration.

Introduction

The objective of this study was to fabricate, characterize, and evaluate two-dimensional (2D) graphene field-effect transistors (GFETs) integrated with polydimethylsiloxane (PDMS) microfluidic channels for highly sensitive pH detection. Graphene, an allotrope of carbon consisting of a single layer of sp^2-hybridized atoms arranged in a honeycomb lattice, exhibits a unique zero-bandgap semimetallic behavior [1]. The tight-binding Hamiltonian of its p_z orbitals results in a linear, energy-momentum dispersion relation near the Brillouin zone corners, where the conduction and valence bands intersect at the Dirac points [1]. Consequently, charge carriers behave as massless Dirac fermions, yielding an extreme ambipolar electric field effect, exceptionally low low-energy scattering rates, and near-ballistic transport at room temperature with intrinsic carrier mobilities exceeding 100,000 $cm^2/V \cdot s$. Furthermore, the density of states vanishes linearly at the Dirac point, making its Fermi level highly sensitive to external electrostatic perturbations.

Leveraging these extreme properties, graphene has become a foundational material for several advanced device architectures:

- **High-Frequency RF Transistors:** The exceptionally high Fermi velocity ($v_F \approx 10^6$ m/s) and mobility of graphene enable rapid carrier transit times across sub-micron channel lengths. This makes GFETs highly advantageous for low-noise amplifiers and analog radio-frequency (RF) electronics capable of operating at transit frequencies in the hundreds of gigahertz, bypassing the limitations of traditional silicon CMOS scaling.
- **Flexible Transparent Electrodes:** Due to Pauli blocking and its atomic thickness, monolayer graphene absorbs only $\pi\alpha \approx 2.3\%$ of incident white light. Combined with its high mechanical tensile strength and electrical conductivity, graphene serves as an optimal, flexible replacement for brittle indium tin oxide (ITO) in organic light-emitting diodes (OLEDs) and conformal photovoltaics.
- **Biochemical Sensors:** Graphene provides an ultimate surface-to-volume ratio, as every carbon atom acts as a surface atom directly exposed to the local environment. This structural paradigm allows for the label-free detection of singular molecular adsorption events or minute fluctuations in the local electrostatic environment, making it uniquely suited for pH and specific biomarker sensing [2, 3].

To facilitate controlled analyte delivery to the sensing area, soft lithography was employed to fabricate microfluidic structures. Microfluidics capitalizes on the dominance of viscous forces over inertial forces at the microscale (low Reynolds number), enabling strictly laminar flow. Key applications of this technology include:

- **Point-of-Care (POC) Diagnostics:** Microfluidic architectures leverage capillary action and controlled fluidic mixing to process complex biological fluids, such as whole blood, enabling rapid, low-cost disease diagnostics outside of centralized laboratory environments [5].
- **High-Throughput Droplet Screening:** Utilizing two-phase immiscible flow (e.g., water-in-oil), microfluidic chips can generate thousands of highly uniform, picoliter-volume droplets per second. Each droplet functions as an isolated bioreactor, enabling the simultaneous screening of massive pharmaceutical libraries against target pathogens or cells [6].
- **Organ-on-a-Chip Platforms:** By continually perfusing nutrient media through PDMS microchannels over cultured human cell lines, researchers can accurately replicate in vivo physiological conditions, including fluidic shear stress and concentration gradients, providing highly accurate models for toxicological assays [7].

Theory, Analysis, and Experimental Methods

The fundamental sensing mechanism of the GFET relies on the modulation of the channel's charge carrier density by an interfacial electrostatic potential. When the graphene channel is exposed to an aqueous electrolyte, an electrical double layer (EDL) forms at the solid-liquid interface. The EDL capacitance (C_{EDL}), comprising the compact Stern layer and the diffuse Gouy-Chapman layer, acts in series with the quantum capacitance (C_q) of the graphene. The pH sensitivity is governed by the site-binding model; surface functional groups on the graphene undergo reversible protonation or deprotonation depending on the bulk hydrogen ion (H^+) concentration [3]. This dynamic equilibrium alters the surface potential, effectively acting as a variable liquid-gate bias that electrostatically dopes the channel and induces a measurable lateral shift in the Dirac voltage (V_{Dirac}) within the transfer characteristics [2].

Device fabrication commenced with the preparation of a highly doped silicon substrate featuring a thermally grown silicon dioxide (SiO_2) dielectric layer. To define the source and drain contact geometries, a multi-layer photolithography process was utilized to ensure a clean lift-off. A lift-off resist (LOR 3A) was spin-coated (500 rpm for 5 s, followed by 3000 rpm for 45 s) and baked. To promote subsequent resist adhesion, Surpass 4000 was spin-coated with an isopropyl alcohol (IPA) wash applied during the final 15 seconds. Finally, a layer of S1813 positive photoresist was applied (500 rpm for 5 s, then 3500 rpm for 45 s). The wafer was exposed using a Heidelberg DWL 66+ maskless aligner utilizing direct laser writing. Upon development in AZ 300 MIF for 120 s, the isotropic dissolution of the LOR created a critical undercut profile, preventing the subsequently deposited metal film from coating the resist sidewalls. A physical vapor deposition (PVD) system was then employed to deposit a 5 nm chromium (Cr) adhesion layer, immediately followed by a 40 nm gold (Au) layer. Immersion in a Remover 1165 solvent bath dissolved the underlying LOR, lifting off the unwanted metal.

Following metallization, monolayer graphene grown via catalytic chemical vapor deposition (CVD) on a 25 μm copper (Cu) foil was transferred to the target substrate [4]. PMMA was spin-coated onto the graphene surface to act as a mechanical support scaffold. Rather than bulk chemical etching, an advanced electrochemical delamination process was utilized; the Cu foil acted as the cathode in a 0.05M NaOH solution at 20V, gently liberating the PMMA/graphene stack [4]. The stack was washed in deionized (DI) water, scooped onto the target wafer, aligned to the pre-patterned electrodes, and subjected to a thermal dehydration bake before the PMMA scaffold was stripped in a warm acetone bath [4].

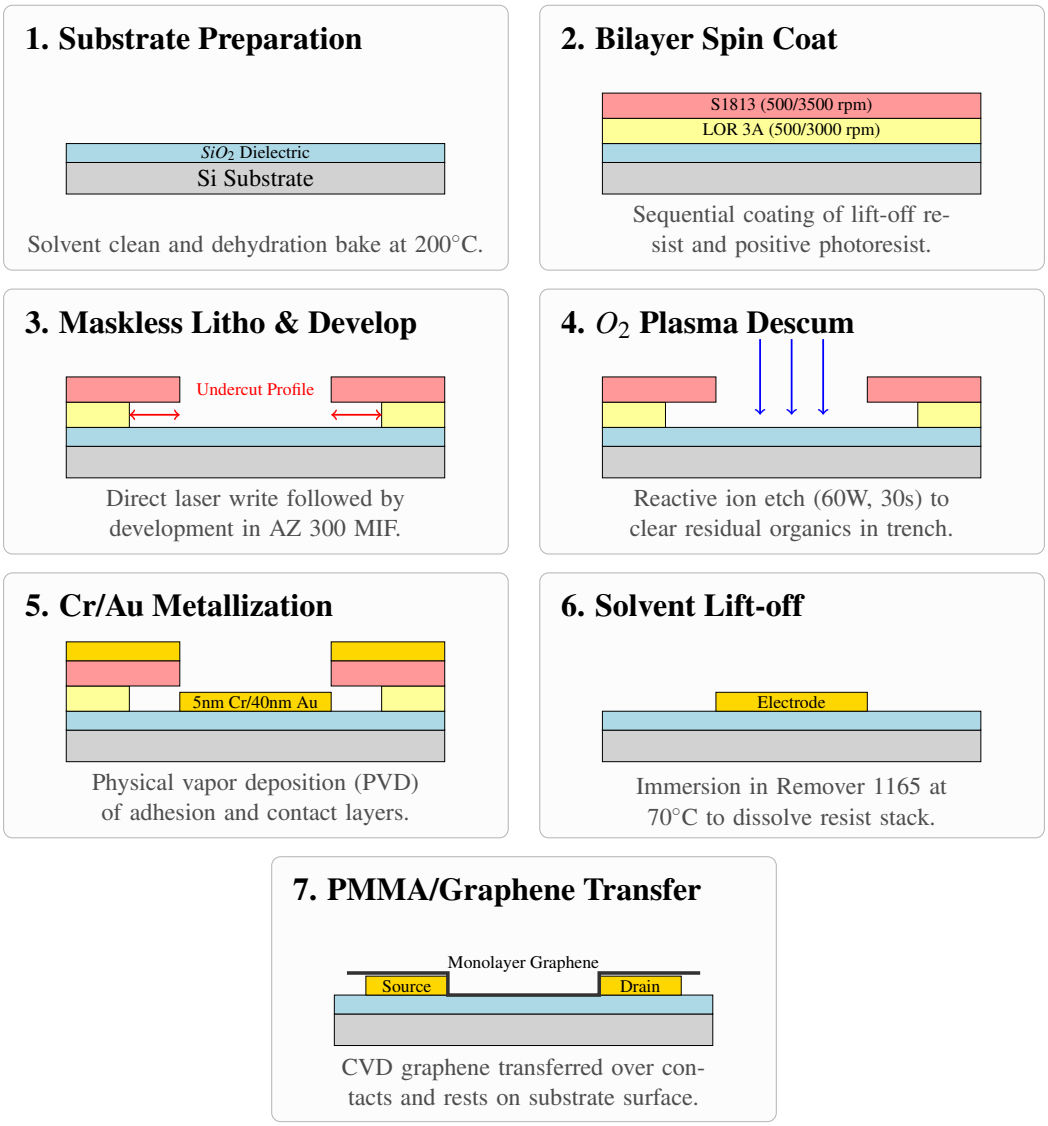

Figure 1: 7-step cross-sectional process flow diagram detailing the device fabrication. The storyboard highlights the critical bilayer undercut and the PMMA-assisted delamination transfer conforming to the substrate surface.

To electrically isolate the individual transistor channels, a secondary photolithography step masked the active regions. The substrate was then subjected to an oxygen (O_2) plasma descum process (flow rate 450, 60 W, 60 s), chemically ashing the exposed carbon lattice to delineate the geometries.

Concurrent with device fabrication, the microfluidic delivery system was manufactured via soft lithography. An SU-8 3050 negative photoresist was spun onto a bare wafer to define a master mold with a targeted 50 μm channel height. A liquid prepolymer of PDMS (Sylgard 184) was mixed at a 10:1 ratio, degassed for 30 minutes, and thermally cured at 100°C. The elastomeric replica was peeled from the master, fluidic ports were punched, and the microchannel was conformally bonded over the GFET array.

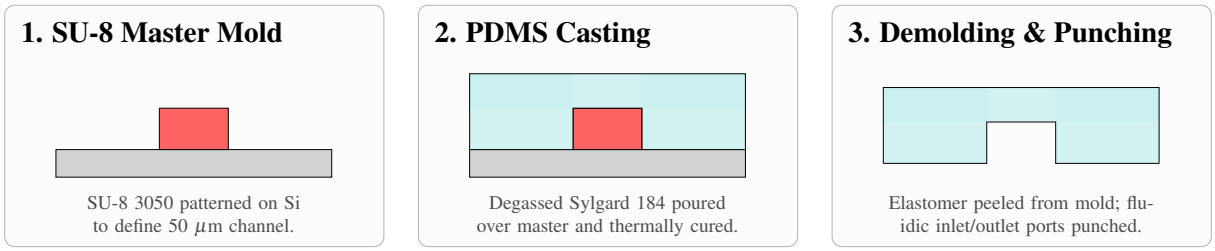

Figure 2: Soft lithography process flow for the PDMS microfluidic delivery system.

Electrical characterization was conducted utilizing a micromanipulator probe station paired with a semiconductor parameter analyzer. Electrochemical pH sensing was evaluated via static drop tests (pH 4, 7, and 10), with thorough DI water rinsing between successive analyte tests.

Results and Discussion

The electrical performance and electrochemical sensitivity of the fabricated GFETs are systematically evaluated. Baseline static electrical transport characteristics are analyzed prior to fluidic integration, followed by dynamic pH response profiling.

The empirical data demonstrates a clear correlation between the physical dimensions of the graphene channel and its maximum electrical resistance, adhering to classical resistance scaling laws. The intrinsic resistance of the two-dimensional graphene sheet is defined by its sheet resistance ($R_s = \rho/T$, where ρ is resistivity and T is thickness) and its geometric aspect ratio (L/W), such that $R_{graphene} = R_s(L/W)$. The total resistance (R_t) measured across the device is modeled as:

$$R_t = 2R_{contact} + R_{graphene} + 2R_{metal}$$

Assuming the resistance of the metal leads (R_{metal}) is negligible, the equation simplifies to a linear function of L:

$$R_t = 2R_{contact} + \frac{R_s}{W}L$$

As illustrated in Figure 3a, the linear relationship confirms that the total resistance is accurately governed by the length of the conductive path, indicating continuous, high-quality graphene sheets free from macroscopic defects. Extrapolating the linear fit to the y-intercept (where $L = 0$ and $R_{graphene} = 0$) yields a total parasitic contact resistance of approximately $500\,\Omega$ ($0.5k\Omega$), resulting in a specific contact resistance of:

$$R_{contact} = \frac{500\,\Omega}{2} = 250\,\Omega$$

Furthermore, the slope of the linear fit ($112\,\Omega/\mu m$, derived from $0.112k\Omega/\mu m$) represents the term R_s/W. Given a constant channel width of $W = 50\,\mu m$, the calculated sheet resistance of the transferred CVD graphene is $R_s = 112 \times 50 = 5600\,\Omega/sq$ (or $5.6k\Omega/sq$).

However, current measured at the Dirac points across varying channel lengths reveals a power-law relationship $I \propto L^{-0.801}$ (Figure 3b), meaning current decreases as the graphene channel gets longer. Given that total resistance is the sum of geometric resistance (R_G) and contact resistance (R_C), at a fixed voltage, current decreases with increasing L. If R_C were negligible, a pure $I \propto L^{-1}$ dependence

would manifest. The exponent of -0.801 rather than -1 confirms that R_C is non-negligible, adding a length-independent offset. This contact resistance becomes relatively more significant at shorter channel lengths, flattening the current–length curve away from the ideal inverse relationship.

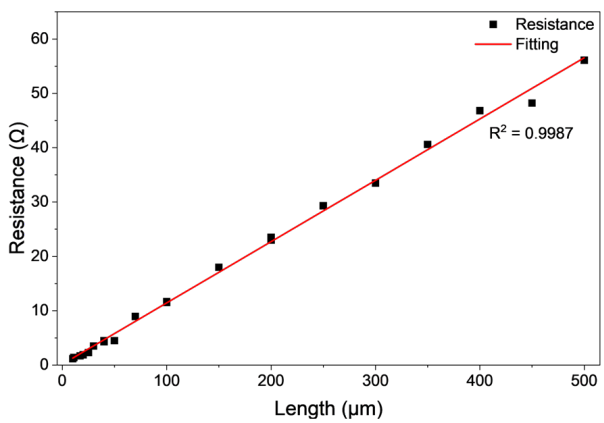

Figure 3: (a) Resistance measured across varying graphene channel lengths, demonstrating linear scaling.

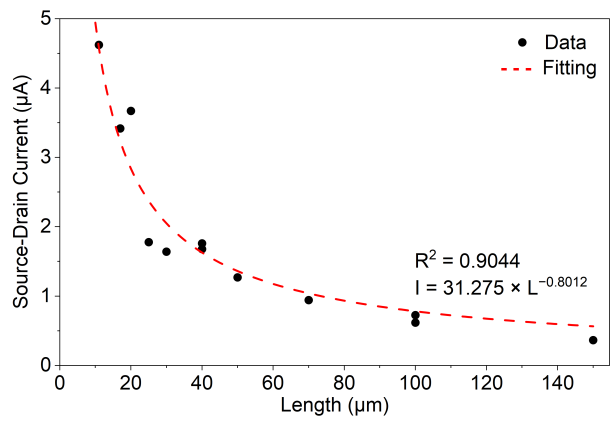

Figure 4: (b) Current measured at Dirac points demonstrating a power-law $I \propto L^{-0.801}$ relationship.

The fundamental field-effect behavior is derived from the gate voltage (V_g) dependent resistance curves, shown in Figure 4a. The most prominent feature is the location of the Dirac point (or charge neutrality point). In an ideal, suspended graphene sheet, charge neutrality occurs exactly at $V_g = 0$ V. The experimental data reveals that the baseline peak resistance for several devices occurs at distinctly positive gate voltages ($V_{Dirac} \approx 39$ V for Device 17, and > 40 V for Device 19). This positive threshold shift indicates native p-doping, attributed to electron-withdrawing adsorbates (oxygen, moisture) and trace residual PMMA acting as electron acceptors. Notably, through process optimization, a champion device (D02) achieved a Dirac point precisely at 0 V. This indicates near-ideal, intrinsic graphene transport where interfacial charge traps and PMMA residues were successfully minimized.

All devices exhibit the hallmark ambipolar transconductance (g_m) profile expected of GFETs (Figure 4b). As V_g is swept from negative to positive values, g_m is initially positive (hole accumulation), reaches a peak, passes through zero at the Dirac point, and becomes negative as electrons become the majority carriers. This sign reversal directly marks the charge neutrality point. Small fluctuations in the g_m curves are indicative of charge inhomogeneity localized electron and hole "puddles" arising from substrate-induced disorder [9, 10].

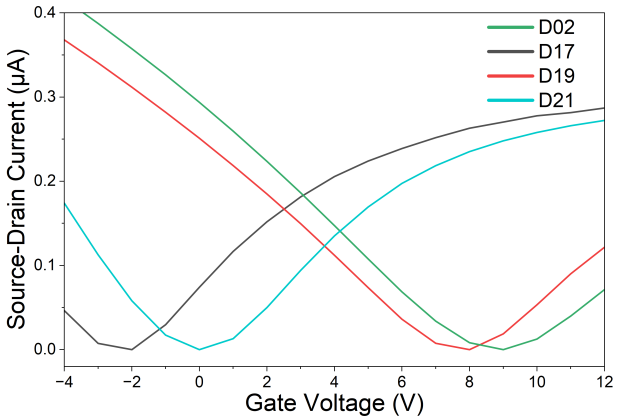

Figure 5: (a) Source-drain current vs gate voltage around the Dirac points (D02, D17, D19, and D21).

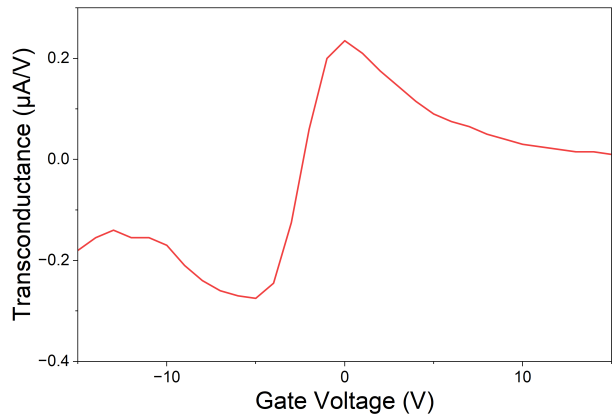

Figure 6: (b) Representative ambipolar transconductance (g_m) curve showcasing charge neutrality.

Following baseline characterization, the GFETs were utilized for electrochemical pH sensing (Figure 5). The devices show increasing resistance with rising pH, consistent with p-type doped graphene behavior. The dynamic transfer shifts manifest as lateral Dirac point shifts, which theoretically follow the Nernst limit of $\sim$ 59 mV/pH at room temperature. Experimentally, the addition of OH^- ions in basic solutions (e.g., pH 10) acts as a local negative gate, donating electrons to the graphene surface, compensating for hole doping, and shifting V_{Dirac} rightward. Conversely, excess H^+ ions (e.g., pH 4) shift V_{Dirac} leftward. Operating in a fixed-gate regime, this lateral voltage shift (ΔV_{Dirac}) translates directly into a measured vertical resistance change (ΔR) [2].

Sensitivity calculations ($\Delta R/\Delta pH$) for 10 μm (D1), 40 μm (D2), and 70 μm (D3) devices yielded 0.005, 0.142, and 0.395 kΩ/pH, respectively. The relatively flat response of D1 is consistent with parasitic contact resistance (R_C) dominating at short channel lengths. When $R_C \gg R_G$, the pH-induced change in the geometric channel resistance is masked by the length-independent contact resistance, severely suppressing sensitivity. Sensitivity is successfully amplified with longer graphene sheets, allowing the pH modulation to dominate the transport metrics.

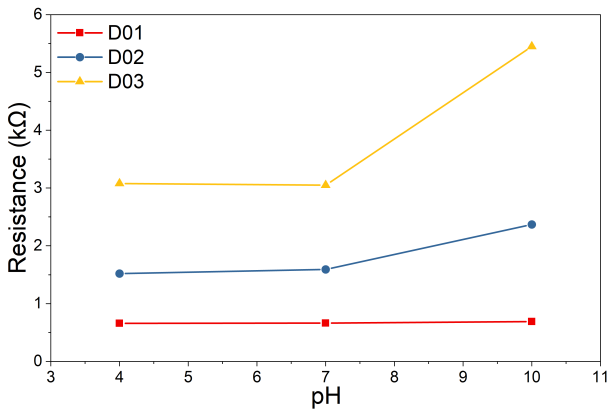

Figure 7: (a) Resistance change across varying pH solutions for Devices D01, D02, and D03.

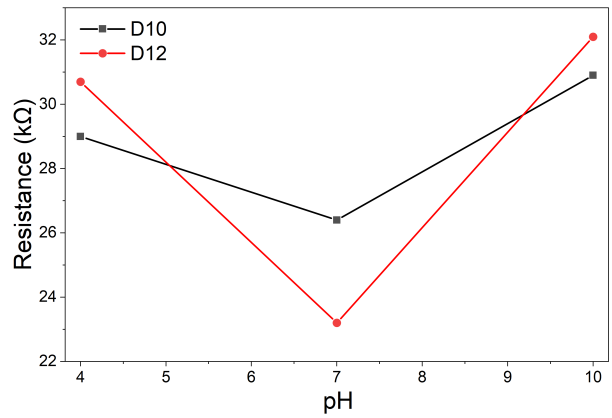

Figure 8: (b) Resistance change across varying pH solutions for Devices D10 and D12.

Conclusions and Recommendations

The fabrication and characterization of GFET devices integrated with PDMS microfluidics successfully demonstrate their capability as highly sensitive pH detectors. The electrical transport measurements confirm the structural integrity of the transferred graphene, evidenced by linear resistance scaling and characteristic ambipolar transport. While baseline devices exhibited native p-doping shifts, process optimization allowed for the isolation of a champion device exhibiting an ideal 0 V Dirac point, confirming the viability of the advanced electrochemical delamination transfer protocol. The systematic modulation of resistance in response to varying pH solutions confirms the site-binding model, wherein interfacial OH^- ions effectively gate the transistor.

Despite the successful demonstration of sensing capabilities, several sources of error and device degradation are identified:

- **Parasitic Contact Resistance:** Transconductance anomalies and power-law current scaling ($I \propto L^{-0.801}$) confirm the presence of significant localized contact resistance (R_C) at the metal-graphene interface. This bottlenecks overall device performance and actively suppresses the pH sensitivity of shorter channel devices.
- **Carrier Scattering:** Trace residual PMMA polymer from the wet transfer process and atmospheric adsorbates act as charged scattering centers, reducing intrinsic carrier mobility and causing charge inhomogeneity (electron-hole puddles).

To mitigate these issues and optimize future device iterations, the following prioritized actions are recommended:

1. **Optimize Contact Topology:** Transition from surface-contacted topologies using titanium to edge-contacted topologies utilizing metals with more favorable work functions (such as palladium or nickel) to minimize the Schottky barrier and heavily reduce parasitic contact resistance, thereby unlocking high sensitivity even in short-channel devices.
2. **Implement High-Temperature Vacuum Annealing:** Prior to fluidic integration, subject the devices to a thermal anneal in a high-vacuum or forming gas (Ar/H_2) environment to aggres-

sively desorb residual PMMA and atmospheric dopants, standardizing the 0 V Dirac point across the entire wafer.

3. **Surface Functionalization and Passivation:** To advance beyond bulk pH sensing toward specific biomarker detection, chemically functionalize the graphene surface with targeted receptor molecules (e.g., aptamers or antibodies) while utilizing a dielectric passivation layer (like atomic layer deposited Al_2O_3) over the metallic contacts to prevent parasitic leakage currents into the fluidic analyte.

References

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